UNIVERSITÉ PARIS 1 PANTHÉON SORBONNE
IAE PARIS
SORBONNE BUSINESS SCHOOL

IDGP ACADEMY

Digital Tools & Benchmarking

International Certificate in Project Management and Organizational Performance

What a platform, an algorithm or a site visit can buy — and the one term in the equation that none of them can

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International Certificate in Project Management and Organizational Performance

17 Oct 2026 · 14:00–17:00 BRT · Online

THIS AFTERNOON

Eight moves, in a deliberate order

01 The digital puzzle

02 What a digital investment buys

03 Philosophy eats AI

04 From a working model to a result

05 Benchmarking: the rules-in-use

06 The transformation vector

07 The Digital Tool Appraisal

08 Building the multiplier

Bring from the morning: your Value Case canvas, and your NUMMI hypothesis with the resolution we reached.

THE CLAIM THE AFTERNOON TESTS



A and P are **additive and purchasable**. C is a **coefficient**.

Technology

A platform enters the bracket. The coefficient is bought separately — or not at all.

Other people's practices

Benchmarking transfers P and purchases A. Neither moves the term that multiplies them.

PART ONE

THE DIGITAL PUZZLE

If this sounds like Fremont with better software, that is the point.

THE SETTING

Two firms. One platform. Everything the same.

FIRM A	THE PURCHASE	FIRM B
✓	Same industry, comparable size	✓
✓	Same enterprise platform, same vendor, same year	✓
✓	Same modules, same implementation partner	✓
✓	Comparable budgets	✓
✓	Same demand-forecasting model, similar history	✓

THREE YEARS LATER

Same purchase. Opposite results.

FIRM A

Materially **shorter lead times**, and a forecast its planners **trust**.

⚠ FIRM B

The platform, the licences, the dashboards — and a planning process that **works around all three**.

Nothing in the purchase differed. Where did the difference come from — and why was it invisible in both business cases?

→ WRITE YOUR HYPOTHESIS IN ONE LINE

PART TWO

WHAT A DIGITAL INVESTMENT ACTUALLY BUYS

Almost every platform is an **A** decision. Some are also **P**. Very few are **C** — and the error is booking the benefit as though C had moved.

THE LANDSCAPE · EXAMPLES ONLY, NOT RECOMMENDATIONS

Six categories — and what each usually buys

Collaboration

Asana, Monday, Trello, Planner, Jira

A — and P if the flow is redesigned

Scheduling & portfolio

MS Project, Primavera P6, Planview, Clarity

A, occasionally P

Analytics & dashboards

Power BI, Tableau, Looker

A — see the caution

Integration & data layer

APIs, integration platforms, warehouse or lakehouse

Often the only genuine P

Automation & workflow

Workflow engines, robotic process automation

P, and A

AI & assistive

Assistants, forecasting models, document generation, agents

A — Parts 3 and 4 say why

Least visible — the data layer — most often carries the real process change.

Most visible — dashboards — change what is seen, not what is done.

THE TRIAGE, APPLIED TO A TOOL

Which term does this tool actually move?

A Assets

Licences, infrastructure, models, dashboards. Something owned that was not owned before.

Test: could a competitor with the same budget hold this position in six months? *Usually yes*.

P Processes

The flow of work changes: approvals reroute, handoffs disappear, a queue is capped.

Test: would the new flow survive if the tool were replaced by another vendor's?

C Capability

The organisation ends up better at improving its processes than it was before.

Test: what can it now do that does not depend on the new system?

⚠ **Training people to operate the system is not capability.** It is part of delivering the asset — and it leaves when they do.

A CAUTION, CARRIED OVER FROM THIS MORNING

Dashboards report variation — not its kind



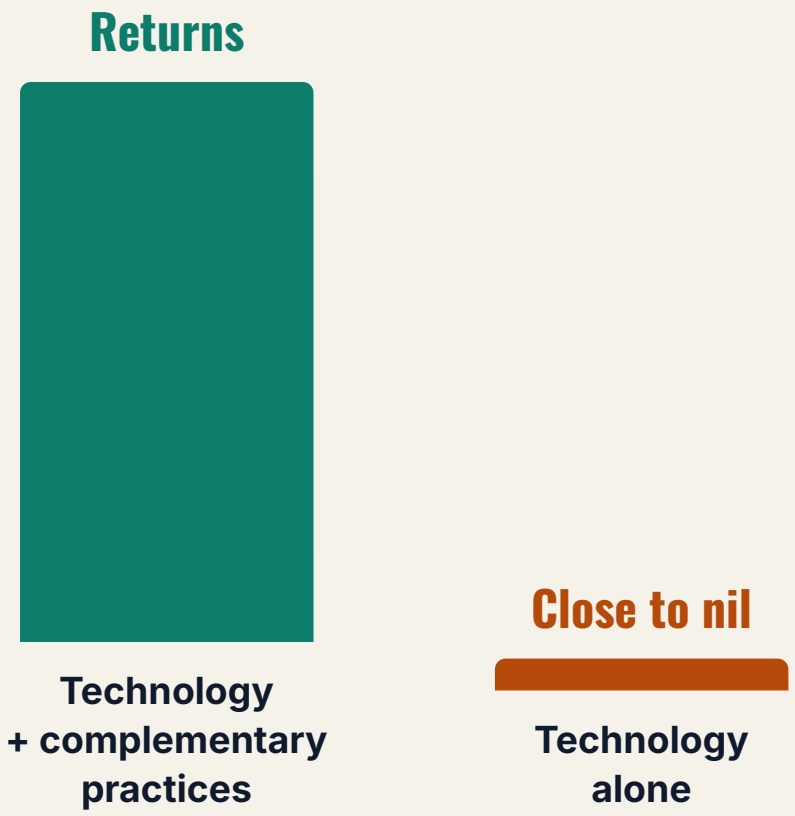
Every point sits **inside the limits**: this is common-cause noise.

A routine that demands an explanation for **every movement** treats noise as a signal — and **makes the process worse**.

The instrument is not the problem. **The routine attached to it is** — a **P** decision that almost never appears in the tooling business case.

THE PRODUCTIVITY PARADOX

Returns appear where the complements exist



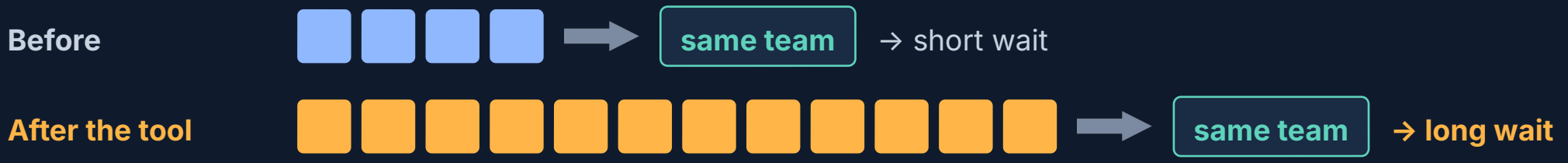
“A platform is a practice that arrives with an invoice attached.”

The pattern survived decades of changing technology — because it was never about the technology.

THE TRAP THAT LOOKS LIKE SPEED

Little's Law does not stop at the screen

Lead time = WIP ÷ Throughput



Easier to start, nothing capped: **more in flight is more time in flight.** The tool is not underperforming — it is performing, and **nobody capped the queue**.

IN PROJECT TERMS

40%

of users on the tool

≠ 40% of the benefit

Every fact needs exactly **one home**.

THE IDEA

A platform’s benefit is not proportional to its adoption.

IN A PROJECT

Used as the working record by 40% of its intended users, a tool delivers close to **none** of a single-source benefit: the other 60% means every figure is still reconciled. A date in two systems is a permanent **reconciliation tax**.

ON MONDAY

List the five facts leadership asks about most. Write where the authoritative version of each lives. Any fact with two homes is your first project — **and it is not a software project**.

PART THREE

PHILOSOPHY EATS AI

The most expensive asset an organisation can buy — and the one whose return depends most completely on a capability that cannot be bought alongside it.

THE BINDING CONSTRAINT IS NOT TECHNICAL

What organisations fund — and what they never write down

INVEST HEAVILY IN

- Algorithms
- Data infrastructure
- Talent

NEVER ARTICULATE

- What the system should value
- How it should know
- What it should represent

Those assumptions do not stay harmless because they were never written. They are executed — at scale, continuously.

Schrage & Kiron, MIT Sloan Management Review, 2025

THREE PILLARS EVERY AI SYSTEM ALREADY HAS

Want · Know · See

☆ Teleology

What the system is told to want — and the ranking among conflicting goals.

A forecast optimised to minimise inventory systematically underestimates seasonal peaks.

Met before: order winners and qualifiers — purposes never ranked.

Epistemology

What it may count as knowing — and how it expresses certainty.

The pandemic invalidated the data behind almost every demand model — and the models kept reporting confidence.

Met before: the andon — a system that knows when it does not know.

Ontology

What it is able to see — which entities and relations exist for it at all.

A sourcing system that sees suppliers as price and quality alone: locally optimal, systemically fragile.

Met before: causal ambiguity — benchmarking the indicator, not the practice.

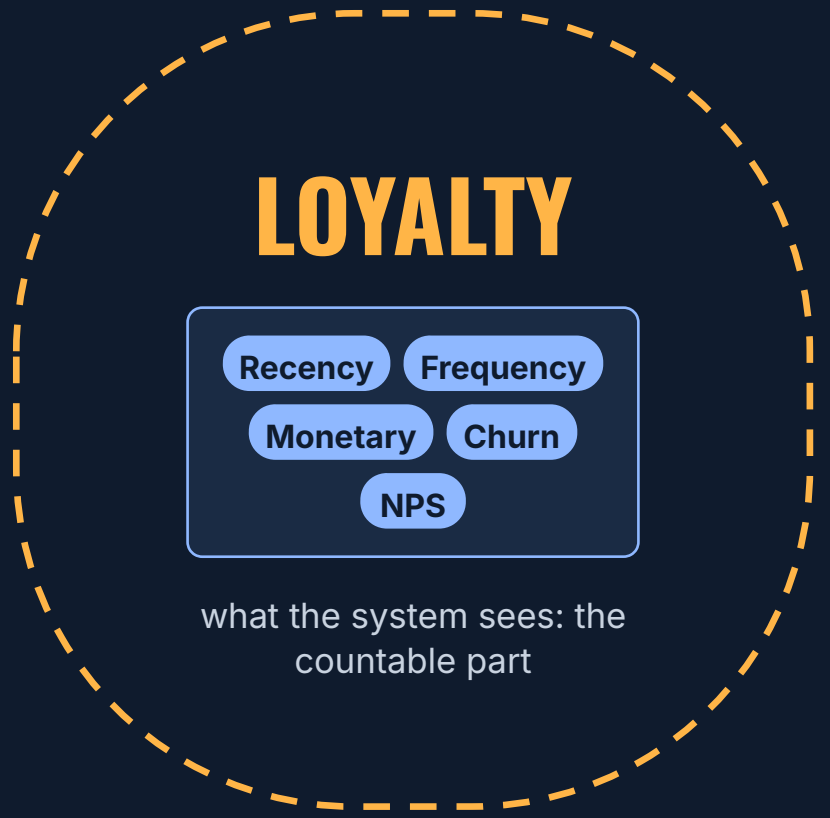
SCHRAGE & KIRON'S OWN EXAMPLE

"The map is not the territory."

— Korzybski, quoted by Schrage and Kiron

Companies want loyal customers — then hand the system only the parts of loyalty that could be counted. It optimises the proxies, faithfully.

Ask of any system: what stand-in has it been given for the thing you actually care about?



THE FAILURE THE THREE PILLARS PREDICT

A trade-off nobody made — decided by default

YEAR 0

A sourcing system is optimised for **cost and delivery**.



YEAR 2

It has systematically **excluded suppliers in the regions the strategy prioritised**.

Not the model

It did exactly what it was asked.

Not a new column

A technical fix just creates debt in the opposite direction.

The real cause

The hierarchy of objectives was never articulated.

Two kinds of debt, in two different places

ACCUMULATES IN A AND P

Technical debt

A **shortcut** in the artefact. A compromise somebody made — and could document.

ACCUMULATES WHERE C SHOULD BE

Philosophical debt

An **absence**. Nobody wrote down the hierarchy of purposes — so nobody can audit it, so nobody can improve it.

A deficit in capability, arriving dressed as a purchase of assets.

FOR THE NEXT STEERING COMMITTEE

Before granting any system autonomy, ask three questions

1

Teleology

Is the hierarchy of purposes **documented** — including what to do when cost, quality and sustainability conflict?

2

Epistemology

Does it **know when it does not know** — and escalate below a confidence threshold?

3

Ontology

Can it recognise a supplier or context it has **never seen** — or only optimise inside its training world?

⚠ **Without those three, autonomy is risk wearing the costume of efficiency.**

PART FOUR

FROM A WORKING MODEL TO AN ORGANISATIONAL RESULT

Part Three was an argument. This is a structure for testing it on a specific implementation.

WHERE THE FRAMEWORK COMES FROM

A systematic review of AI that was actually installed

1,271 → 37

records screened → studies with both an **implementation and a comparison**. Roughly one for every eight that stop at a model or simulation.

Accuracy ≠ result

Where studies produced no effect, predictive performance stayed high wherever it was measured.

PRISMA 2020

Lima, manuscript in preparation. Evidence base: hospitals.

Model accuracy does not separate success from failure. Something else does — and it is not a property of the tool.

EACH NECESSARY, NONE SUFFICIENT

Six conditions from a working tool to a result

THE TOOL

0

Evidence base

Has this class of application produced a measured result in a real organisation?

1

Technical capability

Does it still perform on our data, at the threshold in use?

THE ORGANISATION · C

2

Workflow integration

Does the output reach whoever decides, while the decision is still open?

3

Translation into action

Is there an owner able to act — with the resources to do it?

THE RESULT

4

Outcome for the organisation

Which indicator should move, from which source, against which baseline?

5

Net value

What does the gain cost — and who bears it?

The verdict is never a score. It is the link where the chain breaks — or goes unmeasured.

THE SIX CONDITIONS, IN THE FIELD

What it looks like when each link breaks

Condition	Symptom
0 · Evidence base	Modelling studies and vendor material — no measured implementation anywhere.
1 · Technical capability	Pre-deployment performance not reproduced in routine use.
2 · Workflow integration	Two thirds of the staff never open it — or it arrives after the decision was taken.
3 · Translation into action	Cases correctly identified, and nobody able to act.
4 · Organisational outcome	Benefit stated in general terms, or confirmed only by what users say they felt.
5 · Net value	Gains offset by work displaced to other people or other hours.

Highlighted rows: the two conditions owned by the organisation, not the vendor.

THE TWO MIDDLE CONDITIONS ARE THE ORGANISATION'S

Conditions 2 and 3 are not properties of the tool.

They are capabilities of the organisation that bought it. This is **C, made measurable** — and no vendor can fix it.

Adopting AI is a process-design decision, not a technology purchase.

PART FIVE

BENCHMARKING: DECODING THE RULES-IN-USE

Usually practised as corporate tourism — visit, photograph, buy the software, paste the org chart. That is why it fails. The method is not the reason.

CAMP'S FOUR PHASES

The method, before the critique

1 Plan

Pick a process, not a department. Fix how the measure is defined before anyone looks.

Skipped: comparator chosen for prestige, not comparability.

2 Collect

Same definition on both sides. Record the conditions behind each figure.

Skipped: their figure from a publication; ours calculated differently.

3 Analyse

Size the gap — and, far more important, find **the practice that produces it**.

The fatal one: gap quantified, practice never identified.

4 Adapt

Set targets, change the practice, embed it, re-measure.

Skipped: targets set from the gap; the practice left unchanged.

Most failures are arithmetic before they are strategic.

Same process boundaries on both sides · normalise for scale and mix · record the conditions behind every figure.

CHOOSING THE COMPARATOR

The further away, the harder — and the bigger the payoff

Internal

Another unit of the same firm. Easy access — but the ceiling is the best you already are.

Competitive

A direct rival. Most wanted, least available — and most likely to converge the industry.

Functional

The same function, performed in a different industry.

Generic

An analogous process, unrelated setting. Xerox fixed its warehouse by studying a catalogue retailer.

distance makes the practice visible

Surveys and maturity studies tell you whether a gap exists — never what produces it.

WHY COPY-PASTE GUARANTEES MEDIOCRITY · NUMMI, AGAIN

GM toured NUMMI for twenty years.

Photographed the kanban boards

COPIED · A

Copied the andon lights

COPIED · A

Replicated the workstations

COPIED · P

→ **The other plants remained unproductive and combative.**

The visitors were not looking at the wrong things. They were looking at **the only things that could be looked at**.

THE ARITHMETIC OF FAILED BENCHMARKING

$$V = C \times (A + P) \approx 0$$

rival weak bought copied

Hold this against the hypothesis you wrote at the start of class.

THE EVIDENCE

Why best practice resists transfer

Internal stickiness

Szulanski 1996

Fails even inside one firm — for knowledge reasons: ambiguity and limited absorptive capacity.

→ **Budget for absorption, not only acquisition.**

Causal ambiguity

Lippman & Rumelt 1982

Even the source may not know which elements matter. The knowledge is tacit.

→ **Ask what they stopped doing — and would never give up.**

Interdependence

Rivkin 2000

Partial imitation underperforms no imitation: mismatched elements fight each other.

→ **Identify the bundle, not the practice.**

Time compression

Dierickx & Cool 1989

Accumulated capability cannot be bought quickly — at any price.

→ **Ask how long the reference took. Plan for that.**

THE RULE THIS GIVES YOU

DON’T BENCHMARK THE NUMBER. BENCHMARK THE ROUTINE.

The number has an answer you can put **on a slide**.

The routine — and what you must accumulate to run it — has an answer you can **act on**.

THE STRATEGIC CAVEAT · PORTER

Done well by everyone, benchmarking converges the industry

$VC = (WTP - Cost)_{you} - (WTP - Cost)_{rival}$

A successful imitation is exactly what pulls the rival's bracket up towards yours — and VC towards **zero for everyone**.

Closing a gap on a qualifier

Convergence is the goal. **Benchmark freely — and fund to the threshold.**

Building an order winner

Imitation is the wrong instrument. Ask **what nobody in the set can do yet**.

THE DECODING KEY · SPEAR & BOWEN, 1999

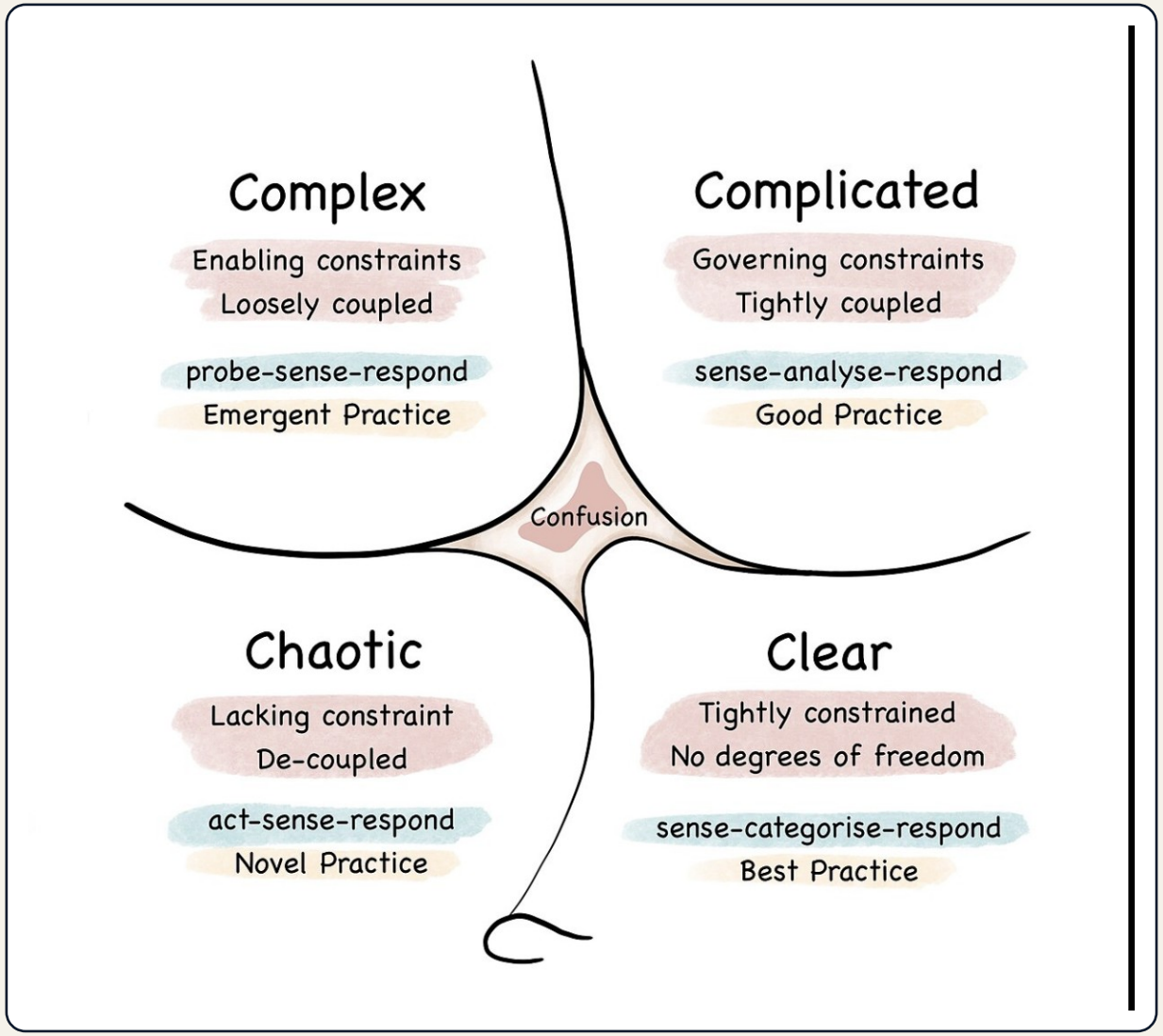
What the visit sees vs. the rule underneath

RULE	NAIVE BENCHMARKING SEES	RIGOROUS BENCHMARKING DECODES
1 · Standardised work	*"They have SOPs and software forms."*	Content, sequence, timing and outcome specified. **Every variance is a testable problem.**
2 · Direct connections	*"They use a central ticketing platform."*	Every internal customer–supplier link is **binary, direct and unambiguous**.
3 · Simple pathways	*"End-to-end routing automated in the ERP."*	One pre-specified path per product, service or signal. **Nothing loops back unless designed to.**
4 · Scientific method	*"They run monthly retrospectives."*	Each improvement is a **refutable hypothesis**, coached by a senior leader, run at the lowest level.

PART SIX

THE TRANSFORMATION VECTOR

If the path cannot be known in advance, what replaces the plan? **Cynefin** says which method the problem admits. **Kata** is the routine that executes it.



CYNEFIN · SNOWDEN

The domain decides the method

Before choosing a tool, a benchmark or a plan, ask which kind of problem this is.

Digital transformation lives in Complex: probe, sense, respond.

CYNEFIN, READ AS A DECISION TABLE

Each domain has its approach — and its trap

Domain	Nature of the problem	Approach	The trap
Simple	Cause and effect obvious to everyone	Sense → Categorise → Respond	Rigidity; blind to a change of context
Complicated	Needs expertise; several right answers	Sense → Analyse → Respond	Paralysis by analysis
Complex	Cause and effect visible only in hindsight — transformation lives here	Probe → Sense → Respond; safe-to-fail experiments	Planning what can only be discovered
Chaotic	No perceptible cause and effect; speed over precision	Act → Sense → Respond	Analysis during a crisis
Confusion	Not knowing which domain you are in	Break it up; reclassify each part	A preferred style, no diagnosis

Leaders are trained for Complicated. In Complex, a detailed plan manufactures a sense of control that delays learning.

2007 · SAME QUESTION, SAME MOMENT, SAME MARKET

Netflix and Blockbuster read the domain differently

Netflix

treated it as COMPLEX

Probe

Streaming as a free extra for DVD subscribers

Sense

Observe who actually adopts it

Respond

Only then pivot the business model

Blockbuster

as SIMPLE

Applied historical best practice to the same question.

Gone by 2010.

Why this indicts benchmarking: pointed at a Complex problem, it imports a **confident answer to somebody else's question** — and the confidence does the damage.

IMPROVEMENT KATA · ROTHER, 2009

The routine that makes probe–sense–respond repeatable

1

Direction or challenge

Aspirational enough to orient; concrete enough to guide daily decisions.

1–3 YEARS

→

2

Grasp the current condition

Go and see the process as it is — and share it, so the next cycle doesn't restart from zero.

CONTINUOUS

→

3

Set the target condition

A measurable future state — neither trivial nor impossible.

UP TO 3 MONTHS

→

4

Run experiments

PDCA, daily. The path is deliberately obscure — if it were clear, there'd be nothing to learn.

WEEKLY / DAILY

⚠ **The commonest Lean failure:** tools adopted, improvement gone in six months. The tools were the content; **the routine was the form** that sustained them.

THE COACHING ROUTINE

Five questions, every time

1. What is the **target** condition?
2. What is the **current** condition?
3. Which **obstacle** are you addressing now?
4. What is your **next step** — and what do you expect?
5. When can we see **what you learned**?

> **The point is not to solve the problem. It is to produce people who solve problems.**
>
> The only form in which capability survives the person who had it. — after Rother, 2009

PUT IT TOGETHER · DISCUSS IN PAIRS

A competitor cuts cost by 30% with generative AI. What do you do?

1 · Cynefin

Simple — copy the technology? Or Complex — discover how AI creates value **for your customer**?

2 · Lean

Is human contact a driver of willingness to pay? Then automating it **cuts cost and revenue** at once.

3 · Kata

A **90-day target condition** and controlled experiments — not a programme whose benefits nobody can state.

The reflex to copy reproduces operational effectiveness — and leaves VC exactly where it was.

IN PROJECT TERMS

A gate asks for

~~a plan~~

a Complex problem can only supply

a hypothesis

Govern it accordingly.

THE IDEA

A gate asks for a plan. A Complex problem can only honestly supply a hypothesis.

IN A PROJECT

Change what an early gate receives: **a target condition and the evidence from the last experiment** — not a schedule and a projection.

ON MONDAY

Rewrite one gate deliverable as: *"By day 90, this measure will read this value; here is what the last four weeks showed."* If the sponsor refuses, the problem is being governed as Complicated.

PART SEVEN · STRUCTURED ACTIVITY

THE DIGITAL TOOL APPRAISAL

Of the digital tooling your project depends on: what is actually being bought — and **what is being left undone**?

THE INSTRUMENT

Four layers and a roadmap — one instrument

VCAP triage	Does it change what we own, how work flows, or what we can improve? Most honest answers land on the first two.	**This morning**
Philosophical debt	AI only: has anyone written down what it should want, may count as knowing, and is able to see?	**Part Three**
Six conditions	Where does the chain from working tool to result break — or go unmeasured? **The maturity read.**	**Part Four**
The reference	What does someone already do that we don't — the routine, not the number — and what must be true here for it to work?	**Part Five**
Maturity & roadmap	Which capability gaps constrain performance most — and in what order can they be closed over six months?	**Conditions read across**

HOW WE RUN IT

Seven sections: five now, two after

1	2	3	4	5	6	7
The tool	VCAP triage	Six conditions	AI audit	Verdict	Reference	Roadmap

In the room · sections 1–5

Groups of three or four. **One scribe** on the fillable instrument. The page carries its own instructions.

After · 6–7

Better done with a week of thought. **No composite score.**

WHAT TO EXPECT

Three honest answers most groups will reach

“Not documented”

Nobody can say what the system optimises for when objectives conflict. **That is the finding** — cheap to fix before deployment, expensive after.

“A — or A and P”

A correctly identified purchase, not a weakness. **Argue the business case on that basis** — not on advantage.

“Not measured”

More often than “no”. Not a failing implementation — **an unobserved one**, a different and more tractable problem.

WHERE THE VERDICT LEADS

A verdict is not a score. It is a located gap.

Gap at **condition 2 or 3** → the problem is Complex	**A 90-day target condition and an experiment — not a requirements document.**
Gap is **a missing measure**	**Measure it — before spending anything further.**

📖 Keep the exported appraisal. With the morning's Value Case, it feeds the module's **Execution Optimization Challenge**.

PART EIGHT

BUILDING THE MULTIPLIER

A can be bought. **P** can be documented and taught. One question is left: **how is C actually built?**

DIERICKX & COOL, 1989

Capability is a stock, not a state

INFLOW

Improvement that produces know-why and know-how together



The valve: psychological safety and credible promises. **Close it and the flow stops.**

C

the capability to improve

A STOCK, NOT A STATE

OUTFLOW

Turnover, disuse, reorganisation



Unregulated — it drains on its own.

A stock adjusts slowly and drains quietly. An organisation good at improving five years ago may no longer be — **and nothing on its balance sheet will say so**.

THE INFLOW, MEASURED

≈ 1 in 4

improvement projects in one factory actually accelerated waste reduction.

The ones producing **know-why and know-how at the same time**. The rest slowed it — or had no effect.

Lapré et al., 2000

WHAT THE INFLOW REQUIRES

Guided experimentation

At the point of execution — the improvement kata in operation.

Team learning

Capability lives in coordinated routines, not in individuals.

Credibility

Never convert improvement gains into redundancies. **Break it once and the inflow stops.**

IN PROJECT TERMS

Lines of a standard changed by your last three retrospectives:

0 ?

If zero: know-why without know-how.
The stock did not move.

THE IDEA

You already own the inflow and the outflow. Few use them as intended.

IN A PROJECT

The **retrospective** is guided experimentation — only if its output changes the standard. The **lessons-learned repository** stores what can be written down: the part that was never the constraint.

THE PMO TEST

After the last three projects closed, what can we now do that we couldn't before — **that doesn't depend on who was on those teams?**

DESIGN IS NOT THE SAME AS PRACTICE

Operational Excellence = Design(VCAP) × Culture

what the organisation specified · multiplied by whether it is actually lived

Design(VCAP)

The blueprint: capability chosen, assets acquired, processes written. **Can be presented, bought, certified and copied.**

Culture

What happens when the auditor has left: the standard followed when inconvenient, **the andon pulled when embarrassing**.

× 0

Excellent processes plus indifferent practice yield **binders**. Where Culture is zero, Operational Excellence is zero — whatever was designed and spent.

WHERE THE DAY ENDS

$$V = C \times (A + P)$$

built bought copied

C is the only term you have to build.

Technology

Buys A, sometimes P. Book the benefit accordingly; put conditions 2 and 3 in scope.

Benchmarking

Transfers the routine, never the number — and converges the industry if all do it well.

Capability

Build it with a routine. Protect the routine with credibility.

MODULE 3 · SESSION 4

THANK YOU

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Reading notes and the Digital Tool Appraisal: tflsixsigma-png.github.io/lean-six-sigma-digital-tools-benchmarking

International Certificate in Project Management and Organizational Performance